Supplementary Material: Plato-Bio

Stefan G. Creadore
Praxa Labs, an independent open-source research initiative, United States

S1. Reproduction commands

```
.venv/bin/python preprint/experiments/run_globin_structure_benchmark.py

.venv/bin/python preprint/experiments/run_globin_structure_benchmark.py \
  --panel-file preprint/experiments/diverse_structure_panel.json \
  --output-dir preprint/results/diverse_structure_benchmark \
  --figures-dir preprint/figures \
  --benchmark-name diverse_structure_panel

.venv/bin/python preprint/experiments/run_temporal_novelty_benchmark.py \
  --fixtures evals/biological_novelty/fixtures/engineering_smoke.json \
  --output-dir preprint/results/temporal_novelty_smoke

.venv/bin/python preprint/experiments/run_temporal_novelty_benchmark.py \
  --fixtures evals/biological_novelty/fixtures/historical_pilot.json \
  --output-dir preprint/results/temporal_novelty_historical_pilot

.venv/bin/python preprint/experiments/run_software_validation.py
```

S2. Original structural input inventory

File	Bytes	SHA-256
1A3N.pdb	461619	1ce17adb5a0a4f986e173aaffa3c3e9910f483f9e32361615a75abb325a1aa7f
3RGK.pdb	147015	5fdba6be63ee6d857f3d31cf639452ade4db527c6b14289a13b6a3aa11660f2a
AF-P02144-F1.pdb	102950	68dd0c9976dd37bc6787b5b398ddc3ac03b6abdb7cc7025298dd503d8bb3c882
AF-P68871-F1.pdb	96551	37b411f5eaa2ff2e6f51774da344f1bd7b54e78c98ce67d9ffb02d8a3dbdf32a
AF-P69905-F1.pdb	92096	d9d7ad4607164de17b65767b467d366909d74f96df5176ee368a9a515aaa1911

The diverse panel contains 29 cached coordinate files; individual hashes and source metadata are in its manifest.

S3. Original globin results

Target	Matched	Identity	RMSD (Å)	Median (Å)	Within 2 Å	ρ	P
hemoglobin_alpha	141	1.000	0.270	0.220	1.000	0.242	0.00388
hemoglobin_beta	145	1.000	0.520	0.453	1.000	0.353	1.34e-05
myoglobin	147	0.987	0.501	0.310	0.993	0.283	0.000515

The original residue-level CSV contains 433 matched positions.

S4. Diverse structural-screen results

Target	Method	Matched	Pred. cov.	Whole RMSD	Core n	Core RMSD	Core 95% interval
hemoglobin_alpha	x-ray diffraction	141	0.993	0.270	141	0.270	0.222–0.285
hemoglobin_beta	x-ray diffraction	145	0.986	0.520	145	0.520	0.446–0.597
myoglobin	x-ray diffraction	147	0.955	0.501	147	0.501	0.360–0.633
lysozyme_c	x-ray diffraction	130	0.878	0.420	130	0.420	0.319–0.528
carbonic_anhydrase_2	x-ray diffraction	257	0.988	0.377	257	0.377	0.318–0.429
kras	x-ray diffraction	166	0.878	3.386	165	2.085	0.868–2.493
cdk2	x-ray diffraction	294	0.987	1.873	263	0.786	0.610–0.878
tp53	x-ray diffraction	194	0.494	6.161	190	3.889	1.079–5.104
estrogen_receptor_alpha	x-ray diffraction	238	0.400	5.326	221	4.961	1.111–6.359
mapk1	x-ray diffraction	337	0.936	1.705	317	0.740	0.488–0.989
superoxide_dismutase_1	x-ray diffraction	153	0.994	0.286	153	0.286	0.185–0.394
transthyretin	x-ray diffraction	116	0.789	0.258	116	0.258	0.206–0.319
cytochrome_c	x-ray diffraction	104	0.990	0.281	104	0.281	0.225–0.305
sumo1	solution nmr	101	1.000	16.610	74	2.576	1.983–3.158
peptidyl_prolyl_isomerase_a	x-ray diffraction	165	1.000	0.531	164	0.331	0.232–0.384

The predeclared rule emitted 27 regions; every row is labeled novelty status not established.

S5. Temporal rediscovery results

Fixture	Condition	Tasks	MRR	R@1	R@10	False novelty
Synthetic	Frequency	5	0.500	0.000	1.000	0.000
Synthetic	TF-IDF	5	1.000	1.000	1.000	0.000
Synthetic	A–B/B–C bridge	5	1.000	1.000	1.000	0.000
Synthetic	Evidence-aware	5	1.000	1.000	1.000	0.000
Historical	Frequency	1	0.333	0.000	1.000	0.000
Historical	TF-IDF	1	0.500	0.000	1.000	0.000
Historical	A–B/B–C bridge	1	1.000	1.000	1.000	0.000
Historical	Evidence-aware	1	1.000	1.000	1.000	0.000

Synthetic results validate engineering behavior only. The historical result is one manually curated retrospective case.

S6. Deterministic validation results

Suite	Tests	Passed	Skipped	Failures	Errors	Wall s
adversarial_safety	57	56	1	0	0	0.86
biology_domain	12	12	0	0	0	1.38
evidence_and_citations	46	46	0	0	0	1.56
full_python	937	931	6	0	0	79.41
genomics_adapters	11	11	0	0	0	1.68

Targeted suites overlap with the full Python suite. Counts are not additive. Skips remain unvalidated live or platform-specific paths.

S7. Measurement-repair acceptance criteria

- A biology GoldenTask constructs Plato with the biology domain.
- Method-signal recall is computed from methods.md and included in summary metrics.
- Evidence JSONL persists drafted Claim rows before EvidenceLink rows.
- A claim with no supporting source remains in the denominator.
- Synthetic rigid transformations are recovered within numerical tolerance.
- Temporal leakage, duplicate sources, prompt-injection quarantine, and abstention are regression tested.

S8. Interpretation boundary

The default evaluator executes idea and method stages only. Live LLM, E2B, Modal, hosted PostgreSQL, and authenticated Hugging Face paths require external credentials or services. Same-model reviewer roles are self-critique, not peer review. The temporal pilot is retrospective, and structural discrepancy candidates are not established novelties.

S9. arXiv packaging notes

The main manuscript is supplied as LaTeX source with embedded figures and tables. Supplemental data are separate. No clinical, human-subject, animal, or identifiable-person data are included. Generative AI assistance is disclosed, and the human author remains accountable. The related Denario arXiv preprint is cited as prior work and is not duplicated by this study.